

Algorithmic appreciation or aversion? The moderating effects of uncertainty on algorithmic decision making

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ABSTRACT

Humans are increasingly making decisions with the aid of algorithms. In some cases, people have exhibited algorithmic aversion, or a tendency to disregard potentially accurate advice from an algorithm. In other cases, the reverse is true, and humans display algorithmic appreciation. Prior work has focused on the role of task type in determining aversion or appreciation, or has considered an individual's agency in the decision making process. In this paper, we posit that certain latent preferences can explain these decisions. We introduce two constructs related to individuals' tolerance for uncertainty and sensitivity to the source of uncertainty and measure them across three different preregistered experimental tasks ($N = 451$ participants total). We find an overall robust tendency towards algorithmic appreciation and find that the measures we introduced significantly moderate the propensity to accept algorithmic advice. We find some heterogeneity across task types and identify circumstances where individuals express aversion instead of appreciation.

CCS CONCEPTS

• **Human-centered computing** → **Empirical studies in HCI**; **Social recommendation**; • **Information systems** → *Expert systems*.

KEYWORDS

Decision making, algorithmic advice, uncertainty, risk preferences, human computer interaction

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1 INTRODUCTION

Individuals regularly make decisions involving uncertainty. The individual making the decision tries to arrive at an optimal solution, despite having imperfect information about potential outcomes. In line with the uncertainty reduction hypothesis [15], people are motivated to reduce subjective uncertainty to achieve certainty about one's environment. One of the ways in which people achieve reductions of uncertainty is by relying upon advice from external sources. Prior literature has examined how individuals incorporate advice when making decisions [14, 28], how advice affects confidence [19, 27, 30], and the performance implications of heeding advice [20, 29, 30].

While advice is often thought of as something provided by humans, decision makers are increasingly receiving input from *algorithmic* sources. This trend is partially due to the superior accuracy of algorithms in many scenarios, as well as the broadening range of tasks that machines can handle such as text generation and generative design [5, 17]. As a result, many organizations are deploying human-machine hybrids – humans making decisions with the "algorithm-in-the-loop" – to tackle complex tasks [12, 32]. These hybrid systems have the potential to outperform a human or algorithm alone [12, 13], however, it is not a given that humans will leverage the input they receive. Thus, maximizing performance requires organizations to identify individuals who are more or less likely to heed algorithmic advice.

In response to this problem, recent research has questioned whether humans exhibit algorithmic aversion or appreciation when presented with advice from a machine. On one hand, there is evidence that people are more willing to accept advice that comes from other humans, particularly when they observe an algorithm err [10]; this phenomenon is referred to as algorithmic aversion. On the other hand, there is evidence that humans are more likely to adjust their decisions when provided with algorithmic advice compared to other humans; this phenomenon is algorithmic appreciation [2, 3, 24]. Yet other studies point to aversion and appreciation as contingent upon the task type and decision-making scenario [7, 9]. A common theme among this stream of research is that appreciation and aversion are framed as *behavioral* indicators, in that humans will express them in certain conditions. Alternatively, we posit that algorithm appreciation or aversion are manifestations of underlying individual differences, i.e., there are latent individual preferences that explain who may be more willing to take algorithmic advice and when. Specifically, we argue that an individual's *tolerance of uncertainty* when making decisions can explain whether or not they will react to algorithmic advice.

Uncertainty is a common element of decision making and is often perceived as aversive [31], and generally takes the form of unknown probabilities with respect to the potential outcomes of a decision [1]. Advice helps the decision maker reduce their uncertainty by updating their prior beliefs about the outcomes. However, prior research suggests that not all individuals tolerate uncertainty in the same way. Some individuals are willing to take greater risks than others given the same information, and this tendency can be linked to distinct neurobiological correlates [8], suggesting that these preferences are linked to individual differences in the processing of uncertainty. Individuals who are more tolerant of uncertainty, i.e., will take greater risks with less information, may thus prefer different types of advice because of their unique decision making processes. Further, research suggests that humans will process uncertainty differently if it comes from social (e.g., other humans) versus non-social (e.g., die roll, computer) sources [4, 11, 21, 22]. Decision makers may thus be more or less willing to accept algorithmic advice, rather than advice from other humans, if they are more or less tolerant of uncertainty from non-social sources. In sum, we posit that these two factors - tolerance of uncertainty and preference regarding uncertainty source - should explain individual tendencies to take algorithmic advice.

We introduce two related individual-difference measures related to uncertainty in decision making. First, we determine an individual's *uncertainty tolerance*, which captures an individual's willingness to make decisions when potential outcomes are unknown. Second, we determine an individual's *uncertainty source preference*, which reflects whether a person is more willing to accept uncertainty when the outcomes are produced by another person or a non-social mechanism. These individual difference measures are computed using an adapted economic lottery choice task to elicit uncertainty preferences [21, 22]. To explore the effect of these underlying preferences, we conduct three preregistered experiments to measure acceptance of algorithmic advice in different task types. We measure individual uncertainty tolerance and uncertainty source preferences for all participants in our experiments using the aforementioned methodology. Our analyses reveal that across task types, uncertainty tolerance and uncertainty source preference significantly explain who will accept algorithmic advice to make decisions.

2 METHODS

To identify the effects of uncertainty preferences on algorithmic advice taking, we conducted three experiments, all of which are preregistered with the Open Science Foundation and approved by the Institutional Review Board at the first author's university. In Figure 1 we provide examples of the lottery choice task (1A), the algorithmic advice tasks (1B), and a schematic of the experimental flow (1C); we proceed to detail each step of the experiments.

2.1 Participant Recruitment

Participants were recruited from Amazon Mechanical Turk¹. We paid subjects a base rate of \$5 USD; to incentivize performance,

¹All participants had to have previously completed at least 500 Human Intelligence Tasks (HITs) with a 98 percent acceptance rate and could not have previously been a subject in an experiment we had conducted. We also required that participants were located in the United States and were not part of the US military.

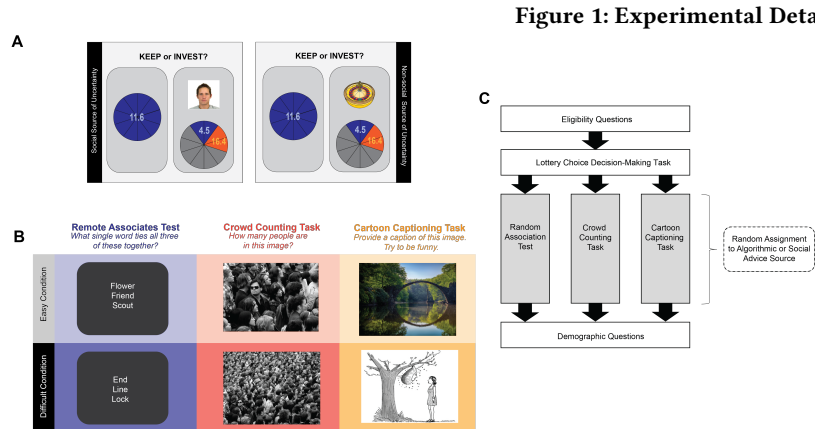
we also informed participants they could earn up to \$4 in bonuses. The task took participants approximately 20 minutes to complete, corresponding to an hourly rate of at least \$15 per hour. We used multiple attention checks, spread throughout the survey. The manipulation check asked subjects whether they received advice from an algorithm or from other people². In total, 451 participants were recruited across the three experiments. For all experiments we gathered demographic data including age, pronouns, number of lifetime HITs completed on MTurk, and income.

2.2 Measurement of Uncertainty Preferences

Uncertainty preferences were measured in this study using an adapted economic lottery choice task [23]. In this task, participants were endowed with 5-15 points and asked to make a total of 88 decisions to invest or not invest. Their goal was to earn as many points as possible by making invest decisions that returned a positive result. They received a bonus payment based on the actual winnings from one of the selected trials. Participants were told that each trial is independent of all other trials and equally likely to be selected for compensation. On each trial, participants were faced with binary decisions to either 1) keep a certain amount which was their allotted endowment or 2) invest their endowed points for the chance of earning more points. For each independent action, participants were told they could choose to invest their points based on information provided by another player (social source) or computer generated chance (non-social source). On their screen, participants were reminded of the source by a picture. Non-social sources were represented visually by a roulette wheel, while the other player was represented using neutral face images from the Chicago Face Database [25]. Overall, half of the trials were social while the remainder were non-social. The potential payoffs and probabilities associated with the payoffs was visually depicted by a pie chart with each piece of the pie corresponding to the probability of 10% (see Figure 1A for an example). Participants were presented with "known" probabilities representing the probability of receiving a high (orange) or low (blue) payoff and "unknown" slices (grey). These "unknown" slices could be associated with the high or low payoff. Uncertainty was defined as the percentage of occlusion (30% or 60%) depicted in the pie chart. These values represent low and high uncertainty, respectively; the numbers were chosen through pilot tests to validate the salience of different uncertainty levels. Uncertainty tolerance was defined as the proportion of decisions to invest; formally, we define the variable as follows: $uncertainty.tolerance_i = \frac{\# investments\ by\ i}{\# opportunities}$. The number of opportunities is equal to 88. By definition, the measure ranges from 0 to 1, and higher values of the measure indicate greater tolerance for uncertainty and vice versa.

Sensitivity to the source of uncertainty was calculated as the difference between the proportion of investments made in the social and non-social conditions. We define this variable as follows: $uncertainty.source_i = \frac{\# investments\ social\ by\ i}{\# opportunities\ social} - \frac{\# investments\ nonsocial\ by\ i}{\# opportunities\ nonsocial}$. The denominator for both fractions is 44, the number of opportunities for investment in the social

²Participants were further excluded from our study if they failed the attention or manipulation checks, or did not enter an MTurk worker ID.



Notes. Details of the experimental procedure. An example instance of the lottery choice task is given in (A). Example prompts for each of the three experimental tasks are shown in (B). The overall schematic of the experiments is shown in (C).

and non-social cases. This measure can range from -1 to 1, where higher values indicated a greater preference for a social relative to a non-social source, and a value of 0 indicates no preference. These measures have been further validated neurologically (see [21, 22]), but those details are beyond the scope of this paper.

2.3 Tasks and Manipulation

All experiments followed the Judge Advisor System (JAS) [14, 28]. In a Judge Advisor System experiment, subjects are asked for their initial answer, then provided with advice, then given the opportunity to answer the question again. We used different forms of decision making to account for potential heterogeneity over tasks. The main manipulation was the same for all experiments - the advice source was labeled as either algorithmic (treatment) or social (control). The manipulation was implemented on a between-subjects basis, with participants randomly assigned to the treatment or control. Within subjects we varied the difficulty of the questions as well as the quality of the advice provided; these manipulations were consistent regardless of the main treatment and were applied in random order. In Figure 1B we provide examples of the three task types used in the study.

In the first study, subjects were asked to work on Remote Associates Test (RAT) problems, which are validated tasks for assessing creative and analytic thinking the domain of language [26]. In a remote associates test problem, subjects are given three words, which are tied together by only one other word. After making an initial guess, participants were provided with advice and the opportunity to submit a new answer. At this point they saw advice based on the treatment condition. The high quality advice was the correct answer based on the RAT database, and the low quality advice was a word that matched exactly two of the three words. Participants repeated this task eighteen times. Our final sample includes 154 individuals and 2,772 observations.

In the second study, we asked subjects to generate a caption for a picture. We presented five images that were cartoons from The New

Yorker's cartoon caption contest (difficult task)³, and five images that were pictures of landscapes (easy task). For the New Yorker cartoons we asked for a funny caption. For the landscapes we asked for an accurate caption. These images were shown to individuals in random order. High quality advice for the humorous images were the winning captions, while the low quality advice was the caption receiving the lowest score. For the landscapes, the authors generated exact descriptions as high quality advice and generic phrases as low quality. Our final sample includes 113 individuals and 1,130 observations.

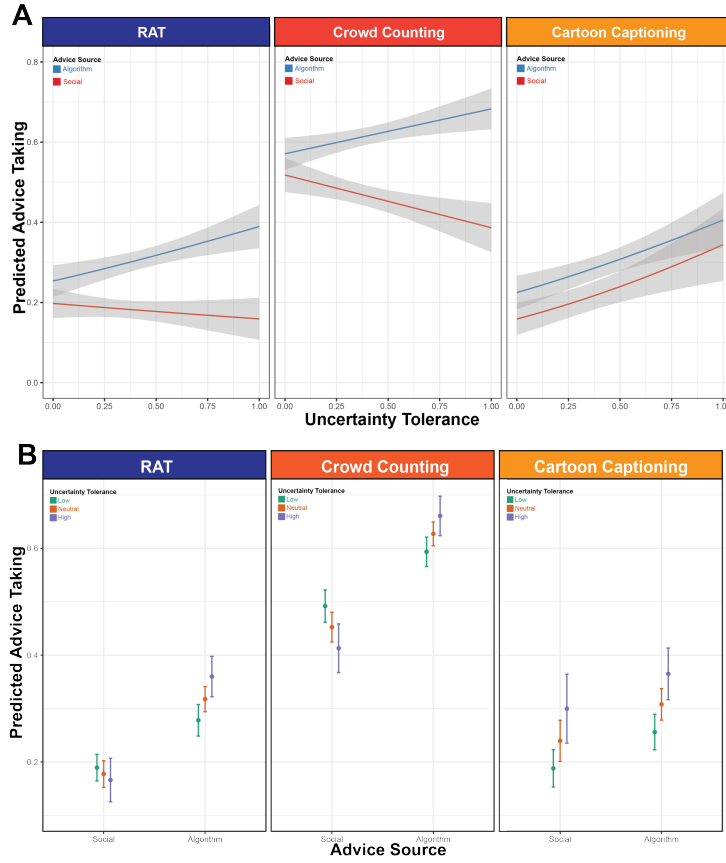
In the third study, we asked subjects to estimate the number of humans in a picture. The images used were taken from previous work with validated answers [16]. Participants saw ten images in random order; five images were considered hard (large number of people in the image), and five images were zoomed-in versions of the hard images. All of the easy pictures had fewer people in them than the hard pictures. High quality advice was the correct number of individuals, while low quality advice was equal to the correct answer plus a random number. Our final sample includes 184 individuals and 1,809 observations (31 observations incomplete)⁴.

2.4 Analytical Approach

To determine the effects of advice source and uncertainty preferences on advice taking, we fit a series of regression models. For the RAT and cartoon captioning tasks, we are testing whether or not the participants changed their answers. Because these are binary outcomes, we employed logistic regression. For the crowd counting task, the dependent variable is weight on advice (WOA), which measures the degree to which a participant makes use of the information they are provided. WOA is defined as: $\frac{|final - initial|}{|advice - initial|}$. Here, *initial* is the numerical answer first submitted by the participant, *advice* is the numerical value provided as advice, and *final* is the

³<https://www.newyorker.com/culture/culture-desk/how-i-became-a-new-yorker-cartoonist>

⁴As a robustness check, we ran the models without individuals who did not finish the task. The results were consistent.

Figure 2: Predicted Algorithmic Advice Taking by Uncertainty Tolerance

Notes. Estimated degree of advice taking across tasks and

by level of uncertainty tolerance. (A) illustrates the moderating effect of uncertainty tolerance. Bands are 95% confidence intervals. (B) provides point estimates at low (0.2), medium (0.5), and high (0.8) levels of uncertainty tolerance by advice source. Levels were determined based on percentiles of responses. Bars are 95% confidence intervals.

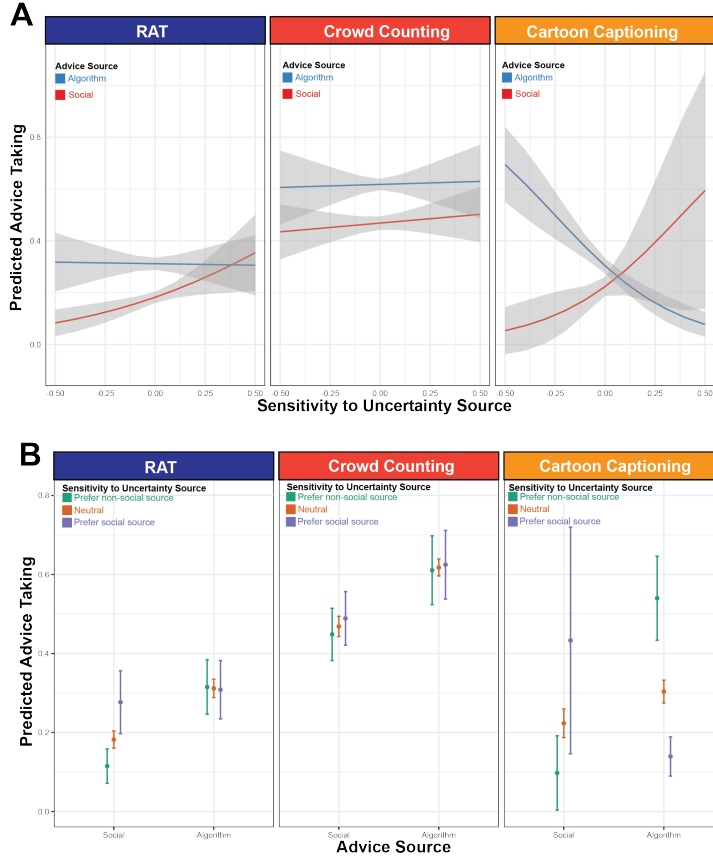
final answer submitted by the participant. In line with prior work, we windorize the WOA measure to be between 0 and 1 [24]. To test the effect of advice source and uncertainty preferences on WOA, we used linear regression. For all models we report robust standard errors clustered on the participant.

3 RESULTS

We illustrate the results in Figures 2 and 3, which show point the moderating effects (A) and point estimates at different levels of the variables across experimental condition (B). In this section all reported values are estimated regression coefficients. We first note that for all three tasks, there is a significant and positive effect of algorithmic advice on whether or not participants changed their answers. Specifically, for the RAT task ($b = 0.757, p < 0.001$) and cartoon captioning task ($b = 0.634, p < 0.001$) participants were 113% and 88.5% more likely to change their answers, respectively, when receiving algorithmic rather than social advice. Further, participants exhibited greater WOA ($b = 0.167, p < 0.001$) when receiving algorithmic advice, which corresponds to 16.7% more change relative to the initial answer for the algorithmic advice condition.

These findings suggest a general tendency towards algorithmic appreciation across three unique task types.

For the RAT task, we find that uncertainty preferences significantly moderate the effect of advice source on the likelihood of changing answers. We find that for individuals with a greater uncertainty tolerance, the effect of algorithmic advice is strengthened ($b = 0.804, p < 0.05$). People with a high tolerance for uncertainty are thus more likely to take advice from algorithms, compared to people with a low tolerance for uncertainty. More precisely, a 10% increase in uncertainty tolerance is associated with an 8.3% increase in the odds of taking algorithmic advice. Further, when individuals are more tolerant of uncertainty from non-social rather than social sources, they are significantly more willing to accept algorithmic advice ($b = -2.386, p < 0.01$). In context, a person who moderately favors non-social sources (-0.05) would be 12.7% more likely to take algorithmic advice than someone who was neutral (0), whereas someone who moderately prefers social sources (0.05) would be 11.3% less likely to take algorithmic advice. In fact, individuals who highly prefer social sources (uncertainty.source > 0.32) will actually be algorithmically averse, i.e., they will be less likely to take advice

Figure 3: Predicted Algorithmic Advice Taking by Uncertainty Source

Notes. Estimated degree of advice taking across tasks and by level of uncertainty source preference. (A) illustrates the moderating effect of uncertainty source. Bands are 95% confidence intervals. (B) provides point estimates for non-social preference (-0.3), neutral preference (0.0), and social preference (0.3) by advice source. Bars are 95% confidence intervals.

from an algorithm than another person. This point can be observed in Figure 3A where the lines cross.

For the cartoon captioning task, we observe that uncertainty preferences have significant effects on advice taking. Unlike with the RAT task, we do not find a significant moderating effect of uncertainty tolerance on advice source. We do find that when individuals are more tolerant of uncertainty from non-social rather than social sources, they are significantly more willing to accept algorithmic advice ($b = -5.103, p < 0.01$). This number implies that a person who moderately favors non-social sources (-0.05) would be 29.1% more likely to take algorithmic advice than someone who was neutral (0), whereas someone who moderately prefers social sources (0.05) would be 22.5% less likely to take algorithmic advice. Further, individuals who only moderately prefer social sources (uncertainty.source > 0.11) will be algorithmically averse. This point can also be observed in Figure 3A where the lines cross.

Finally, for the crowd counting task, we observe that individuals with a higher uncertainty tolerance exhibit significantly higher WOA when receiving advice from algorithms rather than other people ($b = 0.146, p < 0.05$). This value implies that for a 10%

increase in uncertainty tolerance, an individual's WOA will be 0.0146 greater (i.e., 1.46% more weight on advice) when the advice comes from an algorithm, rather than another human. The effect of uncertainty source is negative, as expected, but not statistically significant.

4 DISCUSSION AND CONCLUSIONS

With advancements in artificial intelligence and machine learning technologies, organizations are increasingly leveraging algorithms as advisors for human decision-making tasks. The mainstreaming of this practice calls for fresh theorizing on technology use in decision making. In this paper, we suggest an alternative explanation for why different people exhibit varying levels of algorithmic appreciation or aversion. Namely, we posit that an individual's tolerance to uncertainty and uncertainty source preferences can explain when and why individuals will accept algorithmic advice. Further, we argue that these effects will hold across different types of decision-making scenarios.

We demonstrate a robust algorithmic appreciation effect across three experiments with different task types. Further, we find that

both uncertainty-related variables significantly moderate the advice source treatment effect, though there is heterogeneity across task types. For the most objective task - the crowd counting - individual uncertainty tolerance significantly moderated the treatment effect, such that individuals who were more willing to make decisions with partial information accepted algorithmic advice at a greater rate. By contrast, in the most subjective task - cartoon captioning - it was the uncertainty source preference that significantly moderated the advice source treatment effect. In that experiment, individuals who were more comfortable with non-social sources of uncertainty were much more likely to change their answers when presented with algorithmic advice. Further, individuals who preferred social sources of uncertainty could actually show algorithmic *aversion*. For the RAT task, which had both objective and subjective elements, both variables significantly moderated the advice source treatment such that individuals were more likely to take algorithmic advice if they were more comfortable with uncertainty and preferred non-social to social sources of uncertainty.

Our results have several key implications. First, this study demonstrates that algorithmic aversion or appreciation are not simply rational behaviors in response to situational factors, but are in part manifestations of latent psychological preferences that have well-documented neurobiological correlates [18, 23]. We therefore argue that subjective uncertainty tolerance and sensitivity to the source responsible for the uncertainty explains who does and does not leverage information from algorithms. Second, our results also provide further evidence for algorithmic appreciation, rather than aversion, in tasks that would traditionally be considered subjective. A general assumption is that individuals will favor algorithms when solving problems involving logic or quantitative reasoning, while avoiding algorithms in subjective or open-ended tasks [6, 7, 9]. Based on our findings, we would argue for a more nuanced approach. Our results suggest that individuals will show aversion in subjective tasks (the RAT and Cartoon tasks), but only if they prefer social sources of uncertainty. In other words, people who are more comfortable taking risks when the outcomes are dictated by humans - e.g., the person taking a taxi over the self-driving car - will tend to show algorithmic aversion. By contrast, people who are comfortable with uncertainty, particularly from non-social sources, will generally rely heavily on algorithmic advice, even if that choice is not optimal [2].

From a practical perspective, our study can help inform managers who want to integrate algorithmic advisors into their work groups. Rather than providing the same input to all employees, a more effective strategy might be to tailor the advice to the person receiving it. In other words, someone who prefers a non-social source of uncertainty may be more likely to receive information and feedback when it is given by an algorithmic advisor. By contrast, individuals who are more comfortable with social uncertainty would likely be more receptive to communications in human teams. Additionally, our results suggest that certain individuals will be highly likely to accept algorithmic advice for a variety of decision scenarios, even if the advice is not optimal; in other words, they may be more prone to complacency. Consequently, managers should be wary of blindly implementing algorithmic advisers without training employees to consider advice critically.

In conclusion, we add to the literature on algorithmic aversion and appreciation in human decision making by identifying key individual differences that moderate the propensity to accept algorithmic advice. These measures help us understand who will accept algorithmic advice and when they are likely to do so. We also suggest that the phenomenon of algorithmic aversion may be a manifestation of these subjective cognitive inclinations, rather than a contextual behavior.

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A COMPLETE RESULTS

In each table, robust standard errors clustered on individuals in parentheses. Significance denoted as ⁺p<0.1; *p<0.05; **p<0.01; ***p<0.001.

Table 1: Logistic regression predicting advice taking for RAT task.

	<i>Dependent variable:</i>			
	Changed Answer			
	(1)	(2)	(3)	(4)
<i>Advice.source</i>	0.757*** (0.096)	0.418* (0.164)	0.764*** (0.096)	0.396* (0.164)
<i>Uncertainty.tolerance</i>		-0.256 (0.285)		-0.354 (0.287)
<i>Uncertainty.source</i>			2.156** (0.690)	2.257** (0.698)
<i>Advice.source</i> × <i>Uncertainty.tolerance</i>		0.804* (0.348)		0.899* (0.351)
<i>Advice.source</i> × <i>Uncertainty.source</i>			-2.386** (0.894)	-2.335** (0.905)
Constant	-3.725*** (0.208)	-3.621*** (0.229)	-3.752*** (0.208)	-3.615*** (0.229)
Controls	Included	Included	Included	Included
Observations	2,772	2,772	2,772	2,772
Log Likelihood	-1,445.702	-1,441.568	-1,440.804	-1,436.392
Akaike Inf. Crit.	2,903.404	2,899.136	2,897.608	2,892.784

Table 2: Logistic regression predicting advice taking for cartoon captioning task.

	<i>Dependent variable:</i>			
	Changed Answer			
	(1)	(2)	(3)	(4)
<i>Advice.source</i>	0.634*** (0.136)	0.528* (0.241)	0.562*** (0.140)	0.506* (0.242)
<i>Uncertainty.tolerance</i>		0.618+ (0.326)		0.692* (0.328)
<i>Uncertainty.source</i>			3.542* (1.800)	3.965* (1.822)
<i>Advice.source</i> × <i>Uncertainty.tolerance</i>		0.144 (0.427)		0.014 (0.430)
<i>Advice.source</i> × <i>Uncertainty.source</i>			-5.103** (1.944)	-5.313** (1.967)
Constant	-1.893*** (0.308)	-2.049*** (0.327)	-1.807*** (0.312)	-1.973*** (0.330)
Controls	Included	Included	Included	Included
Observations	1,130	1,130	1,130	1,130
Log Likelihood	-695.268	-689.707	-691.022	-685.640
Akaike Inf. Crit.	1,400.537	1,393.415	1,396.044	1,389.280

Table 3: Linear regression predicting advice taking for crowd counting task.

	<i>Dependent variable:</i>			
	WOA			
	(1)	(2)	(3)	(4)
<i>Advice.source</i>	0.167*** (0.018)	0.105** (0.033)	0.166*** (0.018)	0.105** (0.033)
<i>Uncertainty.tolerance</i>		-0.071 (0.048)		-0.069 (0.048)
<i>Uncertainty.source</i>			0.093 (0.104)	0.086 (0.104)
<i>Advice.source</i> × <i>Uncertainty.tolerance</i>		0.146* (0.064)		0.143* (0.065)
<i>Advice.source</i> × <i>Uncertainty.source</i>			-0.037 (0.180)	-0.065 (0.182)
Constant	0.402*** (0.040)	0.429*** (0.044)	0.403*** (0.040)	0.430*** (0.044)
Controls	Included	Included	Included	Included
Observations	1,809	1,809	1,809	1,809
R ²	0.103	0.106	0.104	0.106